PLANCKS 2026

SOLUTIONS



SOLUTIONS OVERVIEW

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1 Cheerios effect

1. Vertical force balance on a Cheerios gives $F_g + F_b + F_{\gamma,z} = 0$, with the weight of the Cheerios given by

$$F_g = -\frac{4}{3}\pi R^3 \rho_c g,$$

and the buoyancy force by

$$F_b = \rho_m g V_{\text{dis}} = \rho_m g \pi R^3 \left(-\frac{h_c}{R}\sin^2\phi_c + \frac{2}{3} - \cos\phi_c + \frac{1}{3}\cos^3\phi_c \right).$$

The magnitude of the surface tension force is $F_\gamma = 2\pi R\gamma \sin\phi_c$, acting parallel to the liquid-air interface. Hence, its vertical component is given by

$$F_{\gamma,z} = -2\pi R\gamma \sin\phi_c \cos\alpha$$

with α the angle between the liquid-air interface and the negative z-axis. In terms of the interface shape $h(r)$ this angle is given by

$$\cos\alpha = \left.\frac{\mathrm{d}h}{\sqrt{\mathrm{d}r^2 + \mathrm{d}h^2}}\right|_{r=r_c} = -\frac{h_c'}{\sqrt{1+{h_c'}^2}}$$

. Putting all forces together gives

$$\frac{F_g + F_b + F_{\gamma,z}}{2\pi R\gamma} = -\frac{2}{3}BD + \frac{1}{2}B\left(-\frac{h_c}{R}\sin^2\phi_c + \frac{2}{3} - \cos\phi_c + \frac{1}{3}\cos^3\phi_c \right) + \sin\phi_c \frac{h_c'}{\sqrt{1+{h_c'}^2}} = 0, \tag{1}$$

and hence

$$\sin\phi_c \frac{h_c'}{\sqrt{1+{h_c'}^2}} = B\left(\frac{2D-1}{3} + \frac{1}{2}\cos\phi_c - \frac{1}{6}\cos^3\phi_c + \frac{h_c}{2R}\sin^2\phi_c \right).$$

2. If $h_c \ll R$ we can neglect the term that is linear in h_c/R. Furthermore, we find, as $h_c' = \mathcal{O}(h_c/R) \ll 1$,

$$\frac{h_c'}{\sqrt{1+{h_c'}^2}} \approx h_c',$$

and, from trinonometry $\phi_c = \pi - \theta - \arctan h_c' \approx \pi - \theta$ to leading order in the Taylor expansion, such that the equation reduces to

$$\sin(\phi_c) h_c' = B\left(\frac{2D-1}{3} - \frac{1}{2}\cos\theta + \frac{1}{6}\cos^2\theta \right) = B\Sigma.$$

3. The change in gravitational potential energy of Cheerio 1 as it climbs the meniscus of Cheerio 2 is given by the net weight of Cheerio 1 times its rise height. The net weight of the Cheerio 1 is equal to the vertically resolved surface tension force, following the result of (a-b): $2\pi\gamma RB\Sigma$. The vertical displacement along the interface of Cheerio 2 amounts to $h(\ell) = h_c' \sin\phi_c R K_0(\frac{\ell}{L}) = B\Sigma R K_0(\frac{\ell}{L})$ such that

$$E(\ell) = -2\pi\gamma B^2\Sigma^2 R^2 K_0\left(\frac{\ell}{L}\right).$$

4. One Cheerio can lower its gravitational potential energy by climbing the meniscus of its neighbor. As a result, the horizontal interaction force between the two Cheerios is attractive, and given by

$$F(\ell) = -\frac{\mathrm{d}E}{\mathrm{d}\ell} = -2\pi\gamma R^2 B^{5/2}\Sigma^2 K_1\left(\frac{\ell}{L}\right).$$

Given that $K_1(x)$ is positive-valued, the resulting interaction force is negative, implying an attractive force.

5. No, the pushpin and the Cheerio will repel each other. Pushpin is denser than milk, and will hence deform the milk-air interface downwards. To lower its potential energy, the floating Cheerio wants to move upwards, hence away from the pushpin; vice versa, the pushpin wants to lower its potential energy by moving downwards, hence away from the Cheerio. Note that by contrast, two pushpins will again attract each other. Alternative answer: for the pushpin we will have $\Sigma > 1$, while for the Cheerios typically $\Sigma < 1$; the result is a positive $E(\ell)$ and a positive interaction force, meaning repulsion.

End of solution.

2 Partial coherence and 3D imaging

1. If instead mirror A1 has reflectance smaller than 1, we can set $\mathcal{E}_A(t) = r_1 A\mathcal{E}_0(t)$ where r_1 is the *field* reflection coefficient and the output power becomes

$$P_D \propto |A|^2 \left\langle |r_1\mathcal{E}_0(t) + \mathcal{E}(t+\tau)|^2 \right\rangle = |A|^2 \left\langle |\mathcal{E}_0(t)|^2 \right\rangle \left\{ |r_1|^2 + 1 + 2\mathrm{Re}\left[r_1^* g^{(1)}(\tau) \right] \right\}.$$

Assuming for simplicity r_1 to be real, we have, for a Lorentzian source

$$P_D \propto |A|^2 \left\langle |\mathcal{E}_0(t)|^2 \right\rangle \left\{ r_1^2 + 1 + 2r_1\mathrm{Re}\left[g^{(1)}(\tau) \right] \right\} = |A|^2 \left\langle |\mathcal{E}_0(t)|^2 \right\rangle \left\{ r_1^2 + 1 + 2r_1 e^{-|\tau|/\tau_c} \cos(\omega_0\tau) \right\}.$$

We can express it as a function of the mirror position as $P_D \propto \left\{ r_1^2 + 1 + 2r_1 e^{-\frac{2|x|}{c\tau_c}} \cos\left(\omega_0 \frac{2x}{c}\right) \right\}$.

The visibility is given by

$$V \equiv \frac{P_{\max} - P_{\min}}{P_{\max} + P_{\min}} = \frac{(r_1^2 + 1 + 2r_1 e^{-|\tau|/\tau_c}) - (r_1^2 + 1 - 2r_1 e^{-|\tau|/\tau_c})}{(r_1^2 + 1 + 2r_1 e^{-|\tau|/\tau_c}) + (r_1^2 + 1 - 2r_1 e^{-|\tau|/\tau_c})} = \frac{4r_1 e^{-|\tau|/\tau_c}}{2(r_1^2+1)}.$$

A power reflectance of 50% corresponds to a field reflection coefficient $r_1 = \sqrt{0.5} = 0.707$ so that the maximum visibility (around $\tau = 0$) is $V(\tau = 0) = \frac{4r_1}{2(r_1^2+1)} \approx 0.94$. In this calculation we are neglecting the variation of $e^{-|\tau|/\tau_c}$ between the minima and maxima of the fringe pattern, which is well justified as the coherence time is much larger than the period of the fringes.

Note that we can also get to this final result also simply (and without any knowledge of statistical optics) by calculating the minimum and maximum power at $\tau = 0$ as resulting from the constructive and destructive interference of fields:

$$\begin{aligned} P_{\max} &\propto |r_1\mathcal{E}_0 + \mathcal{E}_0|^2 = |\mathcal{E}_0|^2 |r_1 + 1|^2 \\ P_{\min} &\propto |r_1\mathcal{E}_0 - \mathcal{E}_0|^2 = |\mathcal{E}_0|^2 |r_1 - 1|^2 \\ &\Rightarrow V(\tau = 0) = \frac{|r_1+1|^2 - |r_1-1|^2}{|r_1+1|^2 + |r_1-1|^2} = \frac{4r_1}{2(r_1^2+1)}. \end{aligned}$$

Grading: For getting full marks the students need to mention the visibility value and explain how they derived it. Partial marks can be given for attempts at calculating the visibility for this situation

2. In this case, the field reflected from the second mirror A2 can also interfere with the field from B. This will occur only when the displacement of B is such that the corresponding delay matches the delay from A2, so we expect an additional set of interference fringes at $\tau = 2\frac{\Delta y}{c}$. Formally, the power at the detector is given by

$$\begin{aligned} P_D &\propto |A|^2 \left\langle \left| r_1\mathcal{E}_0(t) + t_1^2 r_2 \mathcal{E}_0(1 - 2\frac{\Delta y}{c}) + \mathcal{E}_0(t+\tau) \right|^2 \right\rangle \\ &= |A|^2 \left\langle |\mathcal{E}_0|^2 \right\rangle \left\{ |r_1|^2 + |t_1^2 r_2|^2 + 1 + 2\mathrm{Re}\left[r_1^* \frac{\langle \mathcal{E}_0^*(t)\mathcal{E}_0(t+\tau)\rangle}{\langle |\mathcal{E}_0|^2 \rangle} \right] + 2\mathrm{Re}\left[r_1^* t_1^2 r_2 \frac{\left\langle \mathcal{E}_0^*(t)\mathcal{E}_0(t - 2\frac{\Delta y}{c}) \right\rangle}{\langle |\mathcal{E}_0|^2 \rangle} \right] \right. \\ &\qquad \left. + 2\mathrm{Re}\left[r_1^* t_1^2 r_2 \frac{\left\langle \mathcal{E}_0^*(t)\mathcal{E}_0(t - 2\frac{\Delta y}{c}) \right\rangle}{\langle |\mathcal{E}_0|^2 \rangle} \right] \right\} \\ &= |A|^2 \left\langle |\mathcal{E}_0(t)|^2 \right\rangle \left\{ |r_1|^2 + |t_1^2 r_2|^2 + 1 + 2\mathrm{Re}\left[r_1^* g^{(1)}(\tau) \right] + 2\mathrm{Re}\left[(t_1^2 r_2)^* g^{(1)}\left(\tau + \frac{\Delta y}{c} \right) \right] \right\} \end{aligned}$$

where t_1 is the field transmission coefficient of the first interface, we have neglected multiple reflections bewteen A1 and A2, and used the fact that the reflections from A1 and A2 will not interfere as $2\frac{\Delta y}{c} >> \tau_c$. For the same reason, the

two sets of interference fringes will be well separated, giving rise to the complete interferogram shown in Fig. 1(b). The visibility of the second interferogram will be lower as the light reflected from A2 must traverse A1 twice. We note that this type of interferogram allows measuring the position of multiple backreflectors, and is the basis of optical coherence tomography.

Grading: To get full marks the students should show a sketch closely resembling figure 1(b), and particularly showing two peaks (the second symmetric) spaced by 250 mm and the decay of the fringes with a length scale of few tens of mm. They should also justify their solution.

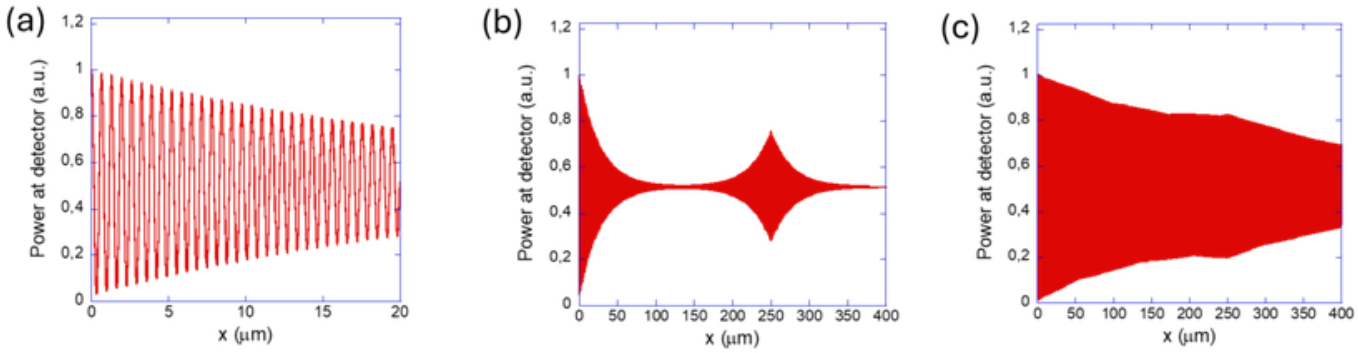


Figure 1: Calculated interferograms for a (a) single and (b) double interface for a source linewidth of 10 nm and (c) double interface for a source linewidth of 1 nm.

3. The insertion of a filter changes the spectrum of the light field, and therefore the coherence time and the interferogram. The position of the filter in the set-up is irrelevant (i.e. it can be placed before of after the BS, as in the figure), as interference occurs in the detector (a way of convincing ourselves of this is to consider that the power spectral density at the detector is the product of the power spectral density of the source and the transmission spectra of the interferometer and of the filter, so that their order is irrelevant). For $\Delta\lambda_{\mathrm{FWHM}} = 1$ nm the coherence time becomes $\tau_c = 1.8$ ps and the corresponding path delay $x_c = \frac{ct_c}{2} = 270$ nm exceeds the distance between the two interfaces, so that the two sets of interference fringes will mix in the interferogram. The resulting interferogram will look as the one shown in figure 1(c).

   Grading: To get the full marks the students should show a sketch resembling the one of figure 1(c), and particularly show that the width of the two peaks is increased with respect to the case (b).

4. The answer to c shows that the resolution is inversely proportional to the spectral width of the source. For higher resolution we need to use a source with a broader spectrum.

   Grading: For getting full marks the students need to mention that a broader spectrum must be used.

5. Single-photon interference produces the same results as classical interference. The photon "interferes with itself" after reflection from multiple surfaces and an interferogram (with the same visibility and x-dependence as calculated for the cases a,b,c) is retrieved when measuring the number of counts as a function of the mirror position.

   Grading: For getting full marks the students need to mention that the same interferograms will be measured.

End of solution.

3 Woodwind instruments

1. We denote the pressure at the package by $p = p_0 + \Delta p$ and the volume of the package by $V = V_0 + \Delta V$, where we split the fluctuations from the mean values. We also denote the displacement of the package by $\boldsymbol{\xi}$. From the adiabatic equation we get in the lowest order approximation

$$\frac{\Delta p}{p_0} = -\gamma \frac{\Delta V}{V_0}. \tag{2}$$

Newton (in lowest order) gives on the other hand

$$m\frac{\partial^2 \boldsymbol{\xi}}{\partial t^2} = -V_0 \nabla p, \tag{3}$$

which gives

$$\rho_0 \frac{\partial^2 \boldsymbol{\xi}}{\partial t^2} = -\nabla p. \tag{4}$$

We get

$$\frac{\partial^2 p}{\partial t^2} = -\gamma p_0 \frac{\partial^2}{\partial t^2}\left(\frac{\Delta V}{V_0}\right) = -\gamma p_0 \frac{\partial^2}{\partial t^2}(\nabla \cdot \boldsymbol{\xi}) \tag{5}$$

$$= \frac{\gamma p_0}{\rho_0}\nabla^2 p, \tag{6}$$

where we used the relation between displacement and volume $\Delta V/V_0 = \nabla \cdot \boldsymbol{\xi}$.

This gives $c = \sqrt{\gamma p_0/\rho_0}$, which in fact, using the ideal gas law, only depends on the temperature and the mass of typical air particles.

Grading: .5 for adiabatic equation to lowest order. .5 for correct formulation of Newton. .5 for the relation between displacement and volume fluctuations .5 for combining ingredients to correct result. Writing down an expression for c is not needed when it is clear from the wave equation.

2. The heat transfer thus scales by $k^2 = \omega^2/c^2$ where k is the wave number and ω is the frequency, while the other term scales by ω. This means for low enough frequencies the heat transfer is negligible.

Grading: .5 for identifying correct factors of k and ω (or equivalent like talking about length and time scales). .5 for using $k = \omega/c$ (or equivalent) to get to the correct conclusion.

3. For the flute and clarinet we have straightforward standing wave solutions $p(t, x) = (A\cos(kx) + B\sin(kx))\cos(\omega t + \phi)$ with frequency $f = \omega/(2\pi) = k\,c/(2\pi)$.

Using the boundary conditions, we get for the flute $A = 0$ and $\sin(kL) = 0$, giving $k = n\pi/L$.

For the clarinet, we get $B = 0$ and $\cos(kL) = 0$, giving $k = (2n-1)\pi/(2L)$.
For the saxophone, one can check that the spherical symmetric wave is given by $p(t, r) = A/r\sin(kr)\cos(\omega t + \phi)$. This can be guessed from the fact that the power is spread evenly, so must decay as r^{-2}, so the pressure fluctuations must decay as r^{-1}. Note that this makes the saxophone indeed closed at the top. The other boundary condition gives $\sin(kL) = 0$, giving $k = n\pi/L$, just like the flute.

Grading: .5 for correct general solution of cylinder wave. .5 for correct general solution of conical wave. .5 per correct result using the boundary conditions (3x).

4. Initially blowing on the reed will give a mix of frequencies, but only the frequencies matching the tube resonance frequencies will create standing waves. At that point the standing waves will create a positive feedback loop, allowing more air when the pressure is already high, while reducing the airflow when the pressure inside is already low, thus amplifying these standing waves.

Grading: .5 for explaining the positive feedback loop. .5 for explaining why only the resonance frequencies get into this loop.

5. Without bell, we get a transmission factor of $1-(Z_{\text{end}}-Z_{\text{tube}})^2/(Z_{\text{end}}+Z_{\text{tube}})^2 \approx 4Z_{\text{end}}/Z_{\text{tube}}$.

   With bell, we have a middle segment with impedance $Z' = Z_{\text{tube}}/2$. This gives a direct transmission factor of

   $$\frac{2Z_{\text{tube}}^2}{(3/2Z_{\text{tube}})^2}\frac{2Z_{\text{end}}Z_{\text{tube}}}{(Z_{\text{tube}}/2)^2} = \frac{64}{9}Z_{\text{end}}/Z_{\text{tube}}, \tag{7}$$

   which is indeed 16/9 as strong.

   Grading: .5 for the correct lowest order result without bell. .5 for multiplying the two transmission factors in the simplified bell case. .5 for getting the right result.

6. The mass of the air gives inertia to flow, so it corresponds to an inductor. To be precise, pressure differences give acceleration, so the corresponding component has a voltage drop proportional to the derivative of the current, which is an inductor.

   There might be other correct explanations. Note that the complex phase of the impedance might suggest that it corresponds to a resistor. This is not true, and due to difference in conventions.

   Grading: .5 for any correct physical explanation.

7. Multiple answers possible. The 'resistor' should have a pressure drop proportional to the amount of flow. Viscous friction between the air and the instrument is a clear example of this.

   The 'capacitor' should have a pressure drop proportional to the integrated flow into the component. An example would be a balloon attached to the end of the instrument: as the balloon fills, more pressure would be created by the rubber of the balloon. In fact, the walls of the tube and closed keys can be seen as capacitors to the outside, albeit with very low capacities.

   Grading: .5 per correct example (2x). .5 bonus point for a very creative *correct* example.

8. The open keyhole corresponds to an inductor. Therefore, for low frequencies it acts as a closed circuit, and thus works as in the ideal case, while for high frequencies, the impedance becomes significant, so more and more transmission past the keyholes occurs.

   Grading: .5 for a correct explanation. If a contestant recognizes the high-pass filter from electronics, without explaining why this is a high-pass filter, this is fine.

End of solution.

4 Gravitational waves of a binary system

1. We have Kepler's third law $\omega^2 = \frac{GM}{a^3}$ (where we have defined $M = m_1 + m_2$) which can be rewritten to $a^4 = \left(\frac{GM}{\omega^2}\right)^{\frac{4}{3}}$. Substituting this gives $\dot{E}_{GW} = \frac{32}{5}\frac{G^{\frac{7}{3}}}{c^5}\mu^2 M^{\frac{4}{3}}\omega^{\frac{10}{3}}$, which can be simplified to $\frac{32}{5}\frac{G^{\frac{7}{3}}}{c^5}(\mathcal{M}\pi f_{GW})^{\frac{10}{3}}$ after some algebra.

2. Assuming isotropic emission, the waves pass through a sphere of radius $4\pi d^2$ such that $F_{GW} = \frac{E_{GW}}{4\pi d^2}$ so $F_{GW} = \frac{8}{5}\frac{(\pi G)^{\frac{7}{3}}}{c^5}(\mathcal{M} f_{GW})^{\frac{10}{3}}$.

3. We have $[E] = \text{kg m}^2\ \text{s}^{-2}$, and we have $[m] = \text{kg}$, $[A] = \text{m}$ and $[f] = \text{s}^{-1}$. From this, we easily see that $E \propto mA^2 f^2$.

4. Two things change when we go from a wave on a string to a wave in spacetime: a gravitational wave propagates through vacuum and the amplitude is dimensionless.

   By analogy with $P \sim$ (medium property)$\times$(amplitude rate)$^2\times$(area), we expect the GW energy *flux* (energy/area/time, $[F] = MT^{-3}$) to take the form

   $$F \sim \kappa\,\langle \dot{h}^2 \rangle, \qquad [\dot{h}^2] = T^{-2},$$

   where $\kappa = G^a c^b$ must supply the missing dimensions:

   $$[\kappa] = \frac{[F]}{[\dot{h}^2]} = \frac{MT^{-3}}{T^{-2}} = MT^{-1}.$$

   With $[G] = M^{-1}L^3T^{-2}$ and $[c] = LT^{-1}$,

   $$G^a c^b: \quad M:\ -a = 1\ \Rightarrow\ a = -1, \qquad L:\ 3a + b = 0\ \Rightarrow\ b = 3, \qquad T:\ -2a - b = -1\ \ (\checkmark).$$

   Hence

   $$\boxed{\kappa \propto \frac{c^3}{G}}.$$

5. For an oscillating strain $h(t) = h_0\cos(\omega_{GW}t)$ with $\omega_{GW} = 2\pi f_{GW}$, $\dot{h} = -\omega_{GW}h_0\sin(\omega_{GW}t)$ and $\langle \dot{h}^2 \rangle = \frac{1}{2}\omega_{GW}^2 h_0^2$. So

   $$F = \frac{c^3}{16\pi G}\cdot\frac{1}{2}\,\omega_{GW}^2 h_0^2 = \frac{c^3\,\omega_{GW}^2}{32\pi G}\,h_0^2.$$

   Equating this to the flux found from the source in Part 2,

   $$\frac{c^3\omega_{GW}^2}{32\pi G}h_0^2 = \frac{8\pi^{7/3}}{5}\frac{(G\mathcal{M})^{10/3}}{c^5 d^2}f_{GW}^{10/3},$$

   and solving for h_0 (using $\omega_{GW} = 2\pi f_{GW}$, so $\omega_{GW}^2 = 4\pi^2 f_{GW}^2$):

   $$h_0^2 = \frac{32\pi G}{c^3\cdot 4\pi^2 f_{GW}^2}\cdot\frac{8\pi^{7/3}}{5}\frac{(G\mathcal{M})^{10/3}}{c^5 d^2}f_{GW}^{10/3} = \frac{64\,\pi^{4/3}}{5}\,\frac{(G\mathcal{M})^{10/3}}{c^8 d^2}\,f_{GW}^{4/3}.$$

   Taking the square root,

   $$\boxed{h_0 = \frac{8}{\sqrt{5}}\,\pi^{2/3}\,\frac{(G\mathcal{M})^{5/3}}{c^4\,d}\,f_{GW}^{2/3}} \qquad \text{i.e.} \qquad h_0\ \propto\ \frac{1}{d}\left(\frac{G\mathcal{M}}{c^2}\right)^{5/3}\left(\frac{\pi f_{GW}}{c}\right)^{2/3}.$$

6.

Source	m_1, m_2 ($M_\odot$)	$\mathcal{M}$ ($M_\odot$)	f_{GW}	d	h_0
GW150914 (BH–BH)	36, 29	28.1	100 Hz	440 Mpc	1.5×10^{-21}
GW170817 (NS–NS)	1.4, 1.3	1.17	100 Hz	40 Mpc	8.4×10^{-23}
White dwarf binary	0.6, 0.6	0.52	2×10^{-3} Hz	10 kpc	6.4×10^{-23}
Massive BH binary	106, 106	92.3	2×10^{-3} Hz	10 Gpc	3.6×10^{-25}

7. The strain is defined by $h = \Delta L/L$, so once h and an (effective) arm length L are known,

$$\Delta L = h\, L.$$

The LIGO/Virgo detectors have physical arms of $L_{phys} = 4$ km (LIGO) or 3 km (Virgo), but Fabry–Perot cavities in each arm make the light bounce back and forth many times, giving an *effective* optical path length of order $L_{eff} \sim 500$ km. Using $h_0 \approx 1.5 \times 10^{-21}$ (GW150914) and $L_{eff} = 500$ km $= 5 \times 10^5$ m:

$$\Delta L = h_0\, L_{eff} \approx 1.5 \times 10^{-21} \times 5 \times 10^5 \text{ m} \approx 7.5 \times 10^{-16} \text{ m} \approx 0.75 \text{ fm}.$$

This is less than the radius of a proton (~ 0.87 fm)! (If instead one naïvely used the bare 4 km physical arm, one would get an even smaller $\Delta L \sim 6 \times 10^{-18}$ m — which is precisely why the folded, resonant optical path is essential: it multiplies the tiny length change by the same large factor by which it multiplies the light's path length.)

Compare this to the wavelength of the laser light used in these interferometers, $\lambda \approx 1064$ nm $= 1.064 \times 10^{-6}$ m (Nd:YAG laser):

$$\frac{\Delta L}{\lambda} \approx \frac{7.5 \times 10^{-16}}{1.06 \times 10^{-6}} \approx 7 \times 10^{-10}.$$

Why this is so hard. The detector must sense a change in the interferometer's arm-length *difference* that is about *a billion times smaller than the wavelength of the light doing the measuring*, and smaller than an atomic nucleus. A conventional interference measurement resolves phase shifts down to perhaps 10^{-4}–10^{-6} of a fringe; getting to 10^{-9}–10^{-10} of a wavelength requires stacking several distinct tricks:

- **Long effective path (resonant cavities):** Fabry–Perot arm cavities (and power/signal recycling) let the light traverse the arm hundreds of times, multiplying the accumulated phase shift (and hence ΔL measured optically) by the same factor that it multiplies the path length, without having to build kilometers-longer physical arms.
- **Enormous circulating laser power:** kilowatts of input power are built up to hundreds of kilowatts circulating in the arms (via power recycling), because the photon-counting (shot-noise) precision of a phase measurement improves as $1/\sqrt{N_{photons}}$.
- **Quantum noise reduction:** injection of squeezed vacuum states of light reduces shot noise below the standard quantum limit in the relevant frequency band.
- **Exquisite isolation from every other source of length change:** multi-stage seismic isolation and pendulum suspensions of the mirrors, ultra-high vacuum beam tubes, and mitigation of thermal (Brownian) noise in the mirror coatings and suspensions — all of which, if left unchecked, would produce spurious length changes many orders of magnitude larger than ΔL.
- **Matched filtering / long integration:** because the signal is a well-modeled, many-cycle chirp, cross-correlating the noisy data stream against theoretical waveform templates (and between the two independent detectors) recovers signals whose instantaneous strain is well below the noise floor of a single measurement.

It is the combination of all of these — not any single trick — that allows LIGO/Virgo to routinely detect length changes a fraction of a femtometer in arms kilometers long, using light with a wavelength of about a micron.

End of solution.

5 The Penrose process

1. The ring is close to the black hole in its ergoregion, so it is forced to co-rotate with the black hole. As a result, the magnet that is connected to the ring with the shaft will also start to rotate and create a time-varying magnetic flux through the coil. This in turn will induce an electric current, thereby lighting the bulb.

2. The energy to light the bulb is extracted from the rotational kinetic energy (angular momentum) of the black hole. As energy is extracted, the black hole's spin parameter a decreases.

3. At a turning point $\dot{r} = 0$, use this to solve for the angular momentum $L^{(0)}$:

$$\dot{r} = 0 \implies L^{(0)} = \frac{-2aM + \sqrt{2Mr_0(r_0^2 + a^2 - 2Mr_0)}}{r_0 - 2M}\mu_0 \,. \tag{8}$$

   Repeat the same for the final particles for which the only change is $E^{(0)} \neq \mu_0$ but $E^{(1,2)}$ and $\mu = \mu_{\rm fin}$:

$$L^{(1,2)} = \frac{-2aE^{(1,2)}M + \sqrt{(E^2_{(1,2)}r_0^2 + \mu^2_{\rm fin}(2Mr_0 - r_0^2))(r_0^2 + a^2 - 2Mr_0)}}{r_0 - 2M} \tag{9}$$

   (in fact, there are two solutions, but this is the only one that has $L^{(0)} > 0$). Now use that angular momentum and energy conservation:

$$L^{(0)} = L^{(1)} + L^{(2)} \tag{10}$$

$$E^{(0)} = \mu_0 = E^{(1)} + E^{(2)} \tag{11}$$

   to solve for $E^{(1,2)}$ by replacing $E^{(2)} = \mu_0 - E^{(1)}$ in the angular momentum conservation law. This yields the desired result after some algebra:

$$E^{(1)} = \frac{\mu_0}{2}\left[1 + \sqrt{\frac{2M}{r_0}\left(1 - \frac{4\mu^2_{fin}}{\mu_0^2}\right)}\right] \tag{12}$$

$$E^{(2)} = \mu_0 - E^{(1)} = \frac{\mu_0}{2}\left[1 - \sqrt{\frac{2M}{r_0}\left(1 - \frac{4\mu^2_{fin}}{\mu_0^2}\right)}\right] . \tag{13}$$

   (Alternatively, use the result and verify that it indeed satisfied Eq. 10-11.)

4. For energy extraction to occur, we require $E^{(1)} > \mu_0$. From the energy equation, this implies that the square root term must be greater than 1:

$$\sqrt{\frac{2M}{r_0}\left(1 - \frac{4\mu^2_{fin}}{\mu_0^2}\right)} > 1 \,. \tag{14}$$

   Solving for r_0 leads to the condition:

$$r_0 < 2M\left(1 - \frac{4\mu^2_{fin}}{\mu_0^2}\right) \,. \tag{15}$$

   Note that this is less than $2M$, which at the equator is the boundary of the ergoregion. By energy conservation, we should have that $E^{(0)} = E^{(1)} + E^{(2)}$. Therefore, if $E^{(1)} > E^{(0)}$, the energy of the second particle $E^{(2)}$ should be negative.[1]

5. When $a = M$, $r_0 = r_+ = M$ so that $E^{(1)} = \mu_0/2(1 + \sqrt{2})$, substituting this into the efficiency equation and using that $E^{(0)} = \mu_0$ yields

$$\eta_{max} = \frac{\sqrt{2} - 1}{2}(\approx 21\%) \,. \tag{16}$$

End of solution.

[1] This negative energy particle is doomed to fall into the horizon, thereby decreasing the black hole mass and angular momentum but in such a way that the irreducible mass of the black hole actually increases!

6 Optical waveguiding principles

1. At the core-cladding interface, total internal reflection must occur. We have Snell's law $n_0 \sin\theta_0 = n_f \sin\theta_f$. TIR occurs if $\theta_f \geq \theta_c$ where the critical angle is $\sin\theta_c = \frac{n_1}{n_f}$. The maximum acceptance angle occurs when the internal ray is at the critical condition $\theta_f = \theta_c$. We have $\sin\theta_f = \sqrt{1-\cos^2\theta_f}$, using geometry $\cos\theta_f = \frac{n_1}{n_f}$ since the angle inside the core is measured relative to the axis, while TIR uses the complementary angle at the interface. So,

$$\sin\theta_f = \sqrt{1-\left(\frac{n_1}{n_f}\right)^2} = \frac{\sqrt{n_f^2 - n_1^2}}{n_f},$$

inserting into Snell's law then gives

$$n_0 \sin\theta_{\max} = \sqrt{n_f^2 - n_1^2}$$

as asked.

2. A ray in the core must satisfy TIR at every interface. For layer k : $\sin\theta_c^{(k)} = \frac{n_k}{n_f}$ so to satisfy all layers $\theta_f \geq \max_k \theta_c^{(k)}$. The strictest condition comes from the largest refractive index outside the core, $n_{\text{clad, max}} = \max\{n_1, n_2, \ldots, n_{N+1}\}$. This layer gives the largest critical angle so the most difficult TIR condition. This gives that

$$\sin\theta_f = \frac{\sqrt{n_f^2 - n_{\text{clad, max}}^2}}{n_f}.$$

Applying Snell's law at the input then gives the result.

End of solution.

PLANCKS Question Quantum Gesa Welker

May 21, 2026

Researchers worldwide are trying to find the limits of Quantum Mechanics, for example by experimentally testing wave function collapse models. We consider a mechanical resonator that is used in such an experiment. It can be described by the Hamiltonian

$$H = \hbar\Omega_m \hat{b}^\dagger \hat{b} + \frac{1}{2}\hbar\Omega_m, \tag{1}$$

where Ω_m is the mechanical resonance frequency of the fundamental resonator mode and $\hat{b}^\dagger$ and $\hat{b}$ are the phonon creation and annihilation operators with

$$\hat{b} = \sqrt{\frac{m\Omega_m}{2\hbar}}\left(\hat{x} + \frac{i}{m\Omega_m}\hat{p}\right) \tag{2}$$

and $\hat{b}^\dagger$ its complex conjugate. m is the resonator mass.

(a) [3 points] Explain qualitatively (in 1-2 sentences) what wave function collapse is. Then name 2 potential causes of wave function collapse. For each of those causes, what are the consequences for the foundations of Physics?

(b) [2 points] For some wave function collapse experiments, the resonator needs to be cooled into its quantum mechancial ground state. Using the equipartition theorem, calculate the effective temperature that the resonator needs to have to be in the quantum-mechanical ground state.

(c) [3 points] Using equation 2, calculate the amplitude of the resonator's zero-point fluctuations. What are the implications of this value for measuring quantum-mechanical superpositions in position?

(d) [2 points] In some collapse models (e.g. the so-called CSL model), wave function collapse will create heat, which shows up as extra noise in the displacement of the resonator. This noise can fluctuate and be different depending on its frequency. Experiments therefore measure the mechanical noise spectrum of the resonator, see Figure 1 as an example. The displacement noise spectral density is defined as the Fourier transform of the displacement autocorrelation

$$S_{xx}(\omega) \equiv \int_{-\infty}^{+\infty} \langle x(t)x(0)\rangle \mathrm{e}^{i\omega t} dt \tag{3}$$

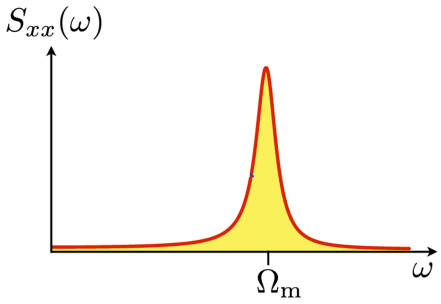


Figure 1: Mechanical noise spectrum of the resonator, graph adapted from arXiv:1303.0733v1

How can you derive the variance of the mechanical displacement, $\langle x^2 \rangle$, from equation 3 and a measured mechanical noise spectrum?

ANSWERS

(a) Wave function collapse is the process where a wave function that is in a superposition of states, reduces to a single state. What wave function collapse actually means, is debated within Physics. The view on the nature of wave function collapse is intrinsically linked to different ways of interpreting quantum mechanics. Potential causes that have been suggested by researchers (2 need to be correctly mentioned):

- Measurement: In the well-known Copenhagen interpretation of quantum mechanics, an observer causes the wave function to collapse during the measurement. As a consequence, quantum physics is non-deterministic. Also, the quantum system doesn't have any real properties before the wave function collapses. In this interpretation, collapse is not a physical process.
- Decoherence: This is not strictly speaking a cause of wave function collapse. It just appears that way, because during decoherence, the system gets entangled with its environment. As a consequence you can treat the wave function as a real thing that doesn't stop existing during collapse. The many-worlds interpretation makes that assumption.
- Gravity: The Diósi-Penrose (DP) model assumes that gravity makes wave functions collapse, since bringing a massive object into superposition means that you also have to superimpose two different spacetimes. If this model was correct, it would have far-reaching conse-

quences on how to unite general relativity and quantum mechanics. Also, in this model, wave function collapse is a physical process.

- Noise sources other than gravity: There are two other important objective collapse models besides DP, namely the Ghirardi-Rimini-Weber (GRW) model and the Continuous spontaneous localization (CSL) model. They assume that some noise source is responsible for the collapse. In these models, wave function collapse is a physical process as well.

(b) Being in the quantum mechanical ground state means that $\hat{N} = \hat{b}^\dagger \hat{b} = 0$. Equipartition states that $\frac{1}{2} k_B T = \frac{1}{2} \hbar \Omega_m$. Therefore, $T = \frac{\hbar \Omega_m}{k_B}$

(c) From equation 2 and the definition of $\hat{b}^\dagger$, we get $\hat{x} = \sqrt{\frac{\hbar}{2m\Omega_m}} \left(\hat{a} + \hat{a}^\dagger\right)$. The amplitude of the zero-point fluctuations is the expectation value of $\hat{x}$ in the $|0\rangle$ state, $x_{zpf} = \sqrt{\langle \hat{x}^2 \rangle}$, with:
$\langle \hat{x}^2 \rangle = \langle 0|\hat{x}^2|0\rangle = \frac{\hbar}{2m\Omega_m} \langle 0| \left(\hat{a}\hat{a} + \hat{a}^\dagger \hat{a}^\dagger + \hat{a}\hat{a}^\dagger + \hat{a}^\dagger \hat{a}\right) |0\rangle = \frac{\hbar}{2m\Omega_m} \langle 1|1\rangle = \frac{\hbar}{2m\Omega_m}$.
Thus, $x_{zpf} = \sqrt{\frac{\hbar}{2m\Omega_m}}$.
When measuring quantum-mechanical superpositions in position, the two superimposed states need to be displaced more than x_{zpf} from each other.

(d) Using Parseval's theorem, we get

$$\int_{-\infty}^{+\infty} S_{xx}(\omega) \frac{d\omega}{2\pi} = \langle x(t)x(0) \rangle. \tag{4}$$

As the autocorrelation at $t = 0$ is the variance, we get

$$\langle x^2 \rangle = \int_{-\infty}^{+\infty} S_{xx}(\omega) \frac{d\omega}{2\pi} \tag{5}$$

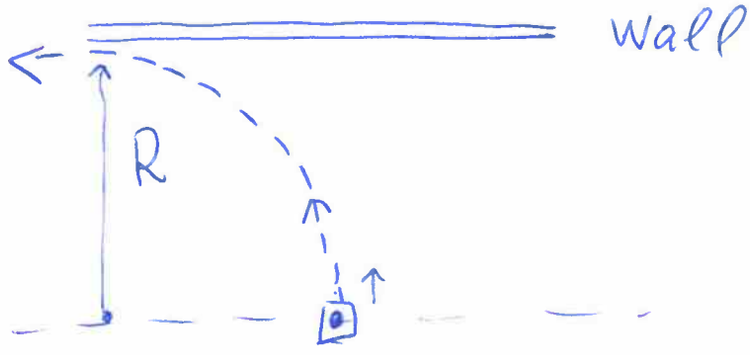
Problem 1

Let us denote m - mass of the system, v_0 - the speed before he noticed the obstacle.

Case 1: Simple break. We apply conservation of energy: $\frac{mv_0^2}{2} = \mu m g L_1$ where L_1 is the "breaking path" - the total path before the full stop.

$$\boxed{L_1 = \frac{v_0^2}{2\mu g}}$$

Case 2:



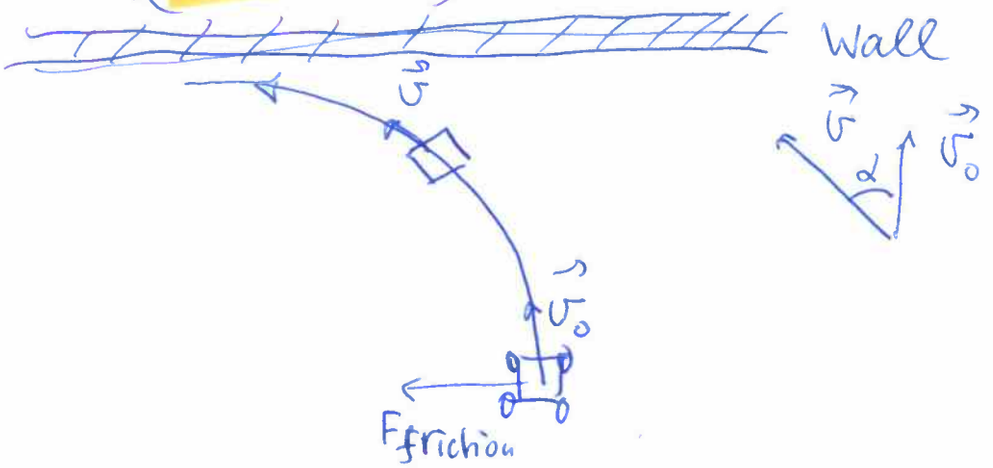
The auto should turn, the path will form a circle. This circle should just "touch" the wall in order to avoid the collision

We apply 2nd Newton law:

$$\frac{mv_0^2}{R} = \mu m g \Rightarrow R = L_2 = \frac{v_0^2}{\mu g}$$

so L_2 is twice less effective!

Case 3 (Problem 1)



Friction is equally distributed between front and rear wheels. This friction is $\perp$ to velocity

Front : 0 0

Rear: The angle between $\vec{F}_{frict}$ and $\vec{v}$ can be changed by breaking and steering!

F_{fr} is maximal when F_{frict} is $\perp$ to the wall!!

In this case the auto approach the wall in the slowest manner. We consider this case now! So,

$\vec{F}_{fr} = \vec{F}_1 + \vec{F}_2$ $\quad \vec{F}_1 \parallel \vec{v}$, $\vec{F}_2 \perp \vec{v}$

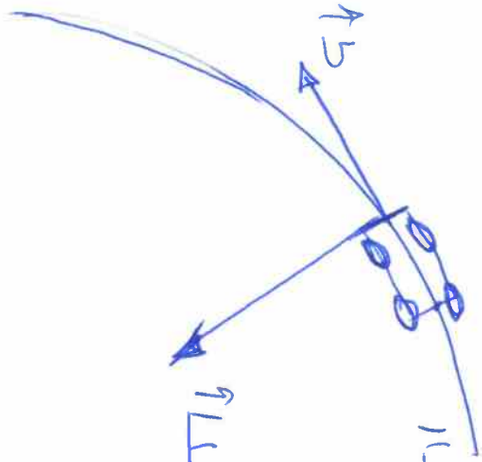
$F_1 = F_{fr}\cos\alpha$; $F_2 = F\sin\alpha$

Explanation

Some ideas ... **Problem 1** Friction force

Case 3

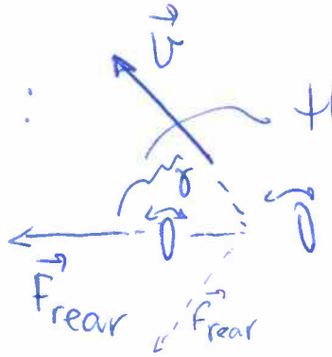
$\vec{F}_{front} \perp \vec{v}$



$\vec{F}_{friction} = \vec{F}_{front} + \vec{F}_{rear}$

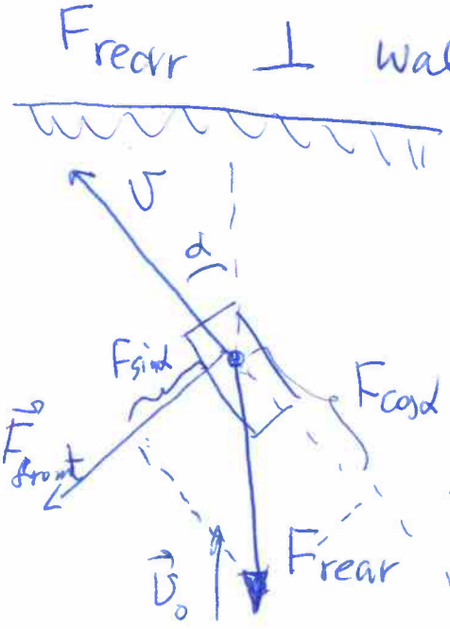
$F \equiv |\vec{F}_{front}| = \mu \frac{mg}{2}$ ($\frac{1}{2} m$ if the mass of ½ auto, for front wheels)

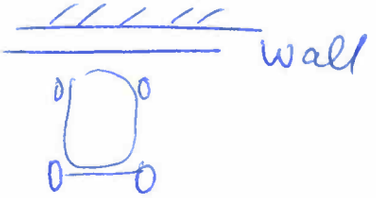
$\vec{F}_{rear}$:



this angle can be changed, by steering and breaking! γ is not constant!

The approaching to the wall will take maximal time if $F_{rear} \perp$ wall :





$$\begin{cases} m\dot{v} = -F_{rear} \cos\alpha \\ \frac{mv^2}{R} = F + F\sin\alpha \end{cases}$$

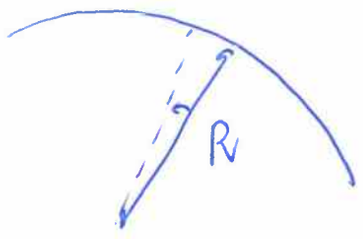
$F = F_{front}$

Problem 1

2nd Newton law:

tangential direction: $m\dot{v} = -F_1 = -F\cos\alpha$ (1)

Normal direction: $\frac{mv^2}{R} = F_2 + F = F(1+\sin\alpha)$ (2)



$v = \dot{\alpha} R$ (3)

From (2) and (3): $mv = F\,\frac{1+\sin\alpha}{\dot{\alpha}}$

Compare with (1):

$$\frac{d}{dt}\left(\frac{1+\sin\alpha}{\dot{\alpha}}\right) = -\cos\alpha$$

$$\frac{d}{d\alpha}\left(\frac{1+\sin\alpha}{\dot{\alpha}}\right)\cdot\frac{d\alpha}{dt} = -\frac{d\sin\alpha}{d\alpha}$$

$$\frac{d}{d\alpha}\left(\dot{\alpha}\,\frac{1+\sin\alpha}{\dot{\alpha}}\right) - \frac{1+\sin\alpha}{\dot{\alpha}}\,\frac{d\dot{\alpha}}{d\alpha} = -\frac{d\sin\alpha}{d\alpha}$$

Or:

$$\frac{d\dot{\alpha}}{\dot{\alpha}} = \frac{2d(\sin\alpha)}{1+\sin\alpha}$$

Problem 1

Initial conditions:

$$\alpha = \alpha_0 = 0$$

$$\dot{\alpha} = \dot{\alpha}_0 \quad , \quad \dot{\alpha}_0 = \frac{v_0}{R_0} \quad ,$$

where R_0 is initial radius of the "swerving":

$$R_0 = \frac{m v_0^2}{F}$$

$$\ln \frac{\dot{\alpha}}{\dot{\alpha}_0} = 2 \ln \frac{1+\sin\alpha}{1+\sin\alpha_0} \quad , \quad \text{or}$$

$$\dot{\alpha} = \frac{v}{R} = \frac{v_0}{R_0} \left(1+\sin\alpha\right)^2$$

Taking into account Eq. (2):

$$v = \frac{v_0 R}{R_0} \left(1+\sin\alpha\right)^2 = \frac{v_0 RF}{m v_0^2} \frac{m v^2}{RF} \left(1+\sin\alpha\right)$$

$$\boxed{v = \frac{v_0}{1+\sin\alpha}}$$

At the nearest to the wall point: $\alpha = \pi/2$,

$$v_{final} = \frac{v_0}{2}$$

To what distance will auto move to the wall?

$$S = \int v \cos\alpha \, dt$$

Problem 1

Because of Eq. (1)

$$\cos\alpha = -\frac{m\dot{v}}{F}$$

$$S = -\int v\dot{v}\frac{m}{F}\,dt = -\frac{m}{F}\int_{v_0}^{v_{final}} v\,dv =$$

$$= -\frac{m}{2F}\left(v_f^2 - v_0^2\right) = \frac{3}{4}\frac{v_0^2}{\mu g} = L_3$$

So, :

$L_1 = \dfrac{v_0^2}{2\mu g}$ – most effective, just break!

$L_2 = \dfrac{v_0^2}{\mu g}$ – twice less effective

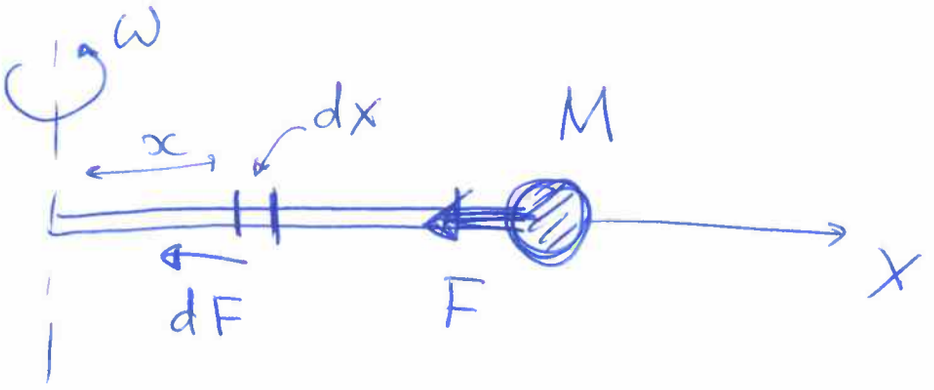
$L_3 = \dfrac{3}{4}\dfrac{v_0^2}{\mu g}$ – Ok... turn and break, still less effective.

1 point – Case 1

2 points – Case 2

7 points – Case 3

Problem 2



2nd law: $M\omega^2 L = F$

deformation is not homogeneous!!

Consider element dx with mass dm. Its undeformed length is dx_0. 2nd law for this element:

$$dm\,\omega^2 x = dF(x) \quad (1)$$

x - its x coordinate, $dF(x)$ - elastic force

$dm = \rho_0 S\,dx_0 = \rho(x) S\,dx$, ρ_0 - density of undeformed cord!

Hook's law: $F(x) = -SE\,\frac{dx - dx_0}{dx_0} = -SE\left(\frac{\rho_0}{\rho(x)} - 1\right)$

Put dm into Eq. 1:

$$\rho(x)\, S\omega^2 x\,dx = E S\rho_0 \frac{d\rho}{\rho^2(x)}, \text{ or}$$

$$(2) \quad x\,dx = A\frac{d\rho}{\rho^3(x)} \quad \text{where } A = \frac{E\rho_0}{\omega^2}$$

Problem 2

Let us integrate Eq. 2 from x to L:

$$x^2 - L^2 = A\left[\frac{1}{\rho^2(L)} - \frac{1}{\rho^2(x)}\right] \quad (3)$$

Let us use $M\omega^2 L = F$ and $F(x) = -SE\left(\frac{\rho_0}{\rho(x)} - 1\right)$ to find $\rho(L)$:

$$1/\rho(L) = 1/\rho_0\left(1 + \frac{M\omega^2 L}{ES}\right) \quad (4)$$

From (3) and (4):

$$\frac{1}{\rho^2(x)} = \frac{1}{A}\left(a^2 - x^2\right) \quad \text{where} \quad a^2 = A\left[\frac{1}{\rho_0} + \frac{ML}{AS}\right]^2 + L^2$$

The mass of cord:

$$m = S\int_0^L \rho(x)\,dx = S\int_0^L \frac{\sqrt{A}\,dx}{\sqrt{a^2 - x^2}} \quad \Rightarrow$$

$$\Rightarrow \rho_0 S L_0 = S\sqrt{A}\,\arcsin L/a \,, \text{ or}$$

$$(5) \quad \sin\frac{\rho_0 L_0}{\sqrt{A}} = \frac{1}{\sqrt{1 + \frac{1}{AL^2}\left(\frac{A}{\rho_0} + \frac{ML}{S}\right)^2}}$$

Problem 2

Equation (5) contains dependence of $L(\omega)$! (because $A \sim \omega^{-2}$)

Let is analyze this.

when $\omega \to 0$ $(A \to \infty)$ $\Rightarrow L_0 \approx L$

Of course! Slow rotation $\to$ no deformation

$L \to \infty$. r.h.s. of Eq.(5) is

$$\left(1 + \frac{M^2}{AS^2}\right)^{-1/2}$$

l.h.s.: Because $\sin\alpha = \left[1 + \frac{1}{\tan^2\alpha}\right]^{-1/2}$

we have: $\tan \frac{\rho_0 L_0}{\sqrt{A}} = \frac{\sqrt{A}\, S}{M}$

Using $A = E\rho_0/\omega^2$:

$$(6) \qquad \tan\left(\omega L_0 \sqrt{\frac{\rho_0}{E}}\right) = S\, \frac{\sqrt{\rho_0 E}}{M\omega}$$

$\omega \to \omega_{critical}$ (solution of Eq. 6) $\Rightarrow L \to \infty$!

This is easier to understand if cord is massless!

Problem 2

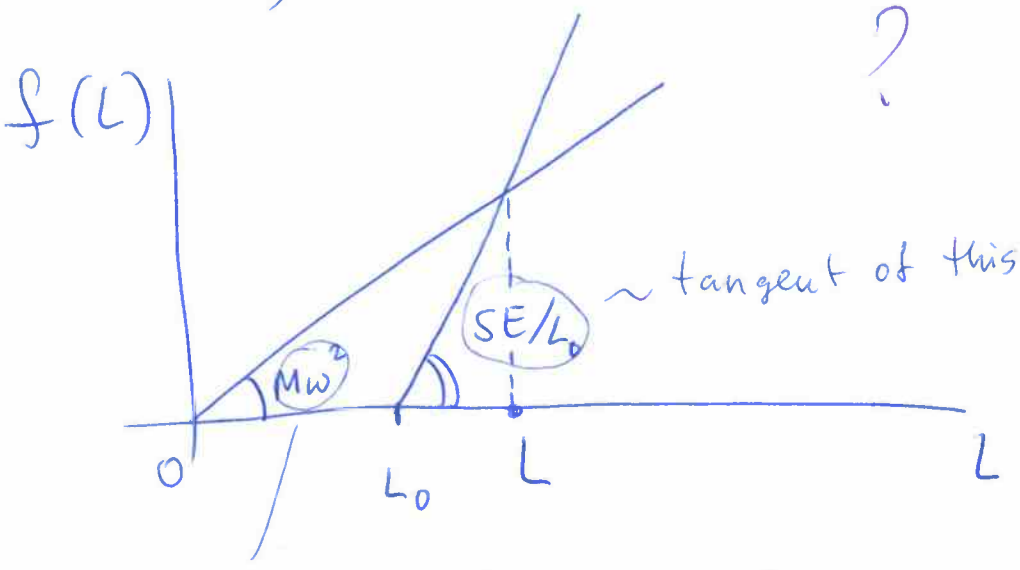
mass cord = 0,

$\rho_0 \to 0$,

$M\omega^2 L_0 = SE$

(7) $M\omega^2 L = \frac{SE}{L_0}(L - L_0) = f(L)$

perhaps? not necessary?

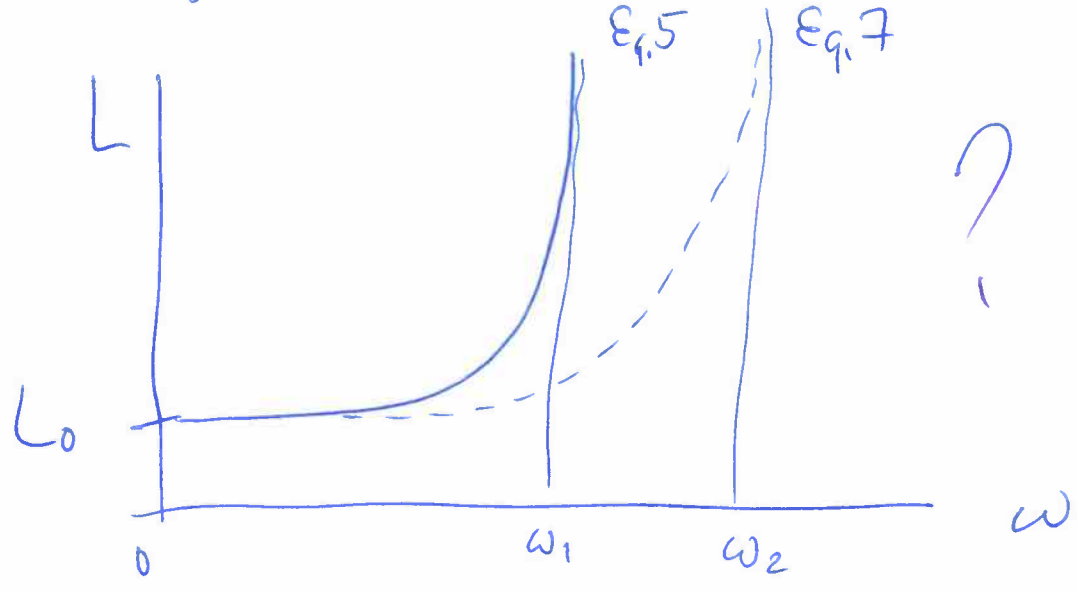


tangent of this angle.

?

L(ω) can be produced using MATLAB....

using Eq. 5 and Eq. 7 :



?

Points: ?? derivation of Eq. 5 – 7 points

Analysis of Eq. 5 – 3 points